



HR MANAGEMENT & WORKFORCE ANALYTICS

06-06-2026

*Submitted By
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CERTIFICATE

This is to certify that the project entitled "**HR Management & Workforce Analytics**" submitted by **Sagar Kumar** in partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Computer Science & Engineering is a bonafide record of work carried out under our supervision and guidance. The project satisfies the academic requirements prescribed by the university and demonstrates practical implementation of modern software engineering practices using ASP.NET Core MVC, SQL Server, Docker, CI/CD pipelines, and Microsoft Azure Cloud.

Guide Signature: _____
Head of Department: _____
Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, department faculty members, and institution for their continuous support and guidance throughout the development of this project.

I would also like to thank Microsoft for providing ASP.NET Core, SQL Server, Azure Cloud services, Docker technologies, and GitHub for enabling modern software development practices.

Finally, I extend my gratitude to my family and friends for their encouragement and support during the completion of this project.

ABSTRACT

Human Resource Management plays a critical role in organizational success. Traditional HR systems often face challenges in workforce monitoring, employee record management, leave administration, reporting, and performance analysis.

The purpose of this project is to develop a centralized Human Resource Management and Workforce Analytics platform using ASP.NET Core MVC and SQL Server.

The system provides:

- Employee Management
- Leave Management
- Workforce Analytics Dashboard
- JWT Authentication
- Role-Based Authorization
- REST APIs with Swagger Documentation
- Docker Containerization
- GitHub Actions CI/CD Pipeline
- Azure Cloud Deployment

The application improves administrative efficiency, data security, reporting accuracy, and workforce visibility through real-time dashboards and analytics.

Keywords:

HRMS, Workforce Analytics, ASP.NET Core MVC, SQL Server, Azure, Docker, CI/CD, JWT Authentication

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CHAPTER 1: INTRODUCTION

Human Resource Management systems are essential for managing employee information, attendance, leave records, workforce performance, and organizational analytics.

Modern organizations require automated systems capable of handling large volumes of employee data while maintaining security, scalability, and reliability.

The HR Management & Workforce Analytics system has been developed to automate HR operations and provide analytical insights for management decision-making.

Major benefits include:

- Centralized employee records
- Reduced paperwork
- Automated leave processing
- Workforce analytics
- Cloud deployment
- Secure authentication

CHAPTER 2: PROBLEM STATEMENT

2.1 Introduction

Human Resource (HR) departments play a vital role in managing employee information, attendance, leave requests, and workforce performance. Many small and medium-sized organizations still rely on spreadsheets, paper-based records, and manual processes to handle HR activities. These traditional methods are time-consuming, error-prone, and difficult to manage as the organization grows. The lack of automation often results in delays, inaccurate records, security concerns, and poor decision-making. Therefore, there is a need for a centralized and digital HR Management System that can streamline HR operations, improve efficiency, and provide valuable workforce insights.

2.2 Existing System Problems

Organizations that use manual HR processes face several challenges:

1. Manual Record Keeping

Employee information is often stored in spreadsheets or paper files. Maintaining and updating these records manually increases the possibility of data duplication, missing information, and human errors.

Impact: Data inconsistency, reduced accuracy, and increased administrative workload.

2. Paper-Based Leave Requests

Employees typically submit leave applications through paper forms, emails, or verbal communication. Managers may take time to review and approve requests, leading to delays.

Impact: Slow approval process, poor tracking of leave balances, and communication gaps.

3. Limited Reporting Capabilities

Traditional HR systems provide limited access to workforce data and analytics. Generating reports often requires manual compilation of information from multiple sources.

Impact: Delayed reporting and poor management decisions due to lack of real-time insights.

4. Security Risks

Sensitive employee information such as salary details, personal data, and employment records may not be adequately protected in manual systems.

Impact: Unauthorized access, data leakage, and compliance issues.

5. Lack of Workforce Analytics

Manual systems do not provide meaningful analytics related to employee trends, leave patterns, departmental performance, or workforce statistics.

Impact: Reduced visibility into organizational performance and inability to make data-driven decisions.

2.3 Problem Analysis

The existing HR process suffers from inefficiencies due to the absence of automation and centralized data management. HR personnel spend significant time performing repetitive administrative tasks, while managers

lack access to real-time workforce information.

The major issues identified are summarized below:

Issue	Impact
Manual Record Keeping	Data inconsistency and human errors
Paper-Based Leave Requests	Delays in approval process
Limited Reporting	Poor decision-making
Security Risks	Unauthorized access to sensitive data
Lack of Analytics	Reduced workforce visibility

2.4 Need for the Proposed System

To overcome the limitations of traditional HR management methods, a modern web-based solution is required. The proposed **HR Management & Workforce Analytics System** will provide:

- Centralized employee data management
- Online leave request and approval workflow
- Secure authentication and role-based access control
- Real-time dashboards and workforce analytics
- Automated reporting and data visualization
- Improved data security and reliability

The system will reduce manual effort, increase operational efficiency, and help management make informed decisions using accurate and real-time workforce data.

2.5 Problem Statement

The current HR management process relies heavily on manual record

keeping, paper-based workflows, and limited reporting mechanisms, resulting in inefficiencies, data inconsistencies, security risks, and lack of workforce visibility. Therefore, there is a need to develop a secure and centralized **HR Management & Workforce Analytics System** that automates HR operations, enhances data accuracy, improves decision-making, and provides meaningful workforce insights through analytics and reporting.

1. CHAPTER 3: OBJECTIVES

The primary objective of the **HR Management & Workforce Analytics System** is to develop a centralized and secure platform for managing employee information and HR operations efficiently. The system aims to implement secure authentication and role-based access control, automate leave management workflows, and provide workforce analytics through interactive dashboards. Additionally, it supports cloud deployment, Docker containerization, and DevOps practices such as CI/CD automation to improve scalability, reliability, and maintainability.

HR MANAGEMENT & WORKFLOW ANALYTICS

CHAPTER 4: LITERATURE SURVEY

Study of:

- Human Resource Information Systems (HRIS)
- Workforce Analytics Platforms
- Cloud-Based Enterprise Applications
- Microsoft ASP.NET Core Architecture
- DevOps Methodologies

Comparison table:

System	Features	Limitations
BambooHR	Employee Management	Paid
Zoho People	HR Automation	Subscription
Workday	Enterprise Scale	High Cost

CHAPTER 5: PROPOSED SYSTEM

The proposed **HR Management & Workforce Analytics System** is a web-based solution designed to automate and streamline human resource operations within an organization. The system consists of a centralized HR Portal for managing employee records, Leave Management for handling leave applications and approvals, and an Employee Management module for maintaining employee information efficiently. An Analytics Dashboard provides real-time insights and workforce statistics to support informed decision-making. The system also exposes secure REST APIs for integration with external applications and services. Cloud deployment ensures accessibility, scalability, and reliability. The proposed solution offers several benefits, including enhanced security through authentication and authorization, improved scalability, automated workflows, reduced manual effort, and easier system maintenance, making HR operations more efficient and effective.

CHAPTER 6: SYSTEM ARCHITECTURE

Architecture Pattern:

Three-Tier Architecture

Presentation Layer

↓

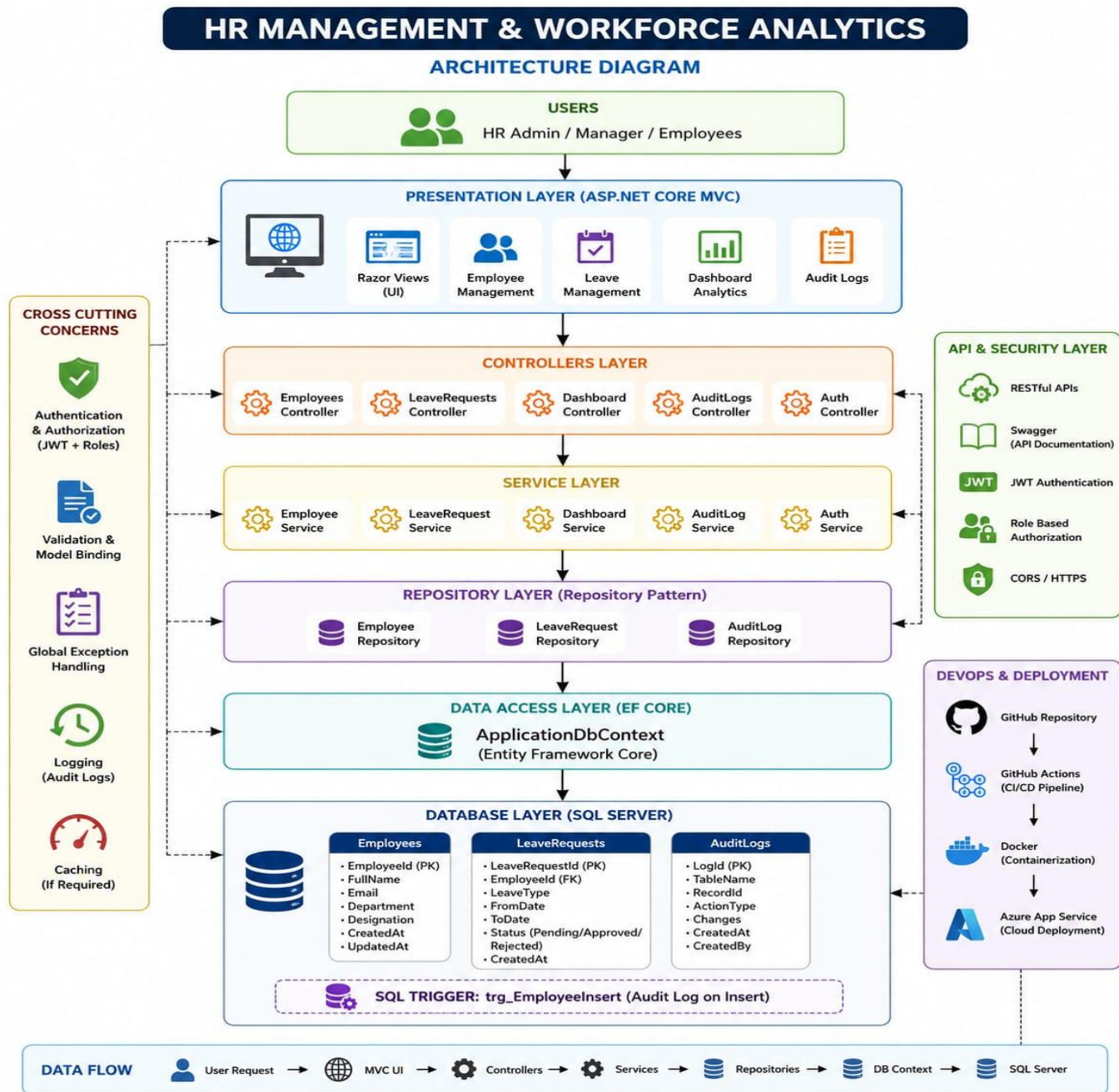
Business Layer

↓

Data Access Layer



SQL Server Database



Architecture Components:

1. ASP.NET Core MVC UI
2. Controllers
3. Services
4. Repository Layer
5. Entity Framework Core
6. SQL Server
7. Azure App Service
8. Docker Container

CHAPTER 7: TECHNOLOGY STACK

The **HR Management & Workforce Analytics System** is developed using modern web technologies to ensure performance, scalability, security, and maintainability. The frontend is built using **HTML5**, **CSS3**, **Bootstrap**, and **JavaScript**, which provide a responsive and user-friendly interface accessible across different devices and screen sizes. The backend is developed using **ASP.NET Core 8 MVC** and **C#**, enabling robust business logic implementation, efficient request handling, and a structured Model-View-Controller architecture.

For data storage and management, **Microsoft SQL Server** is used as the relational database management system. **Entity Framework Core** serves as the Object-Relational Mapping (ORM) framework, simplifying database operations and reducing the need for complex SQL queries. Security is implemented using **JWT (JSON Web Token) Authentication**, ensuring secure user login and role-based access control.

The application is containerized using **Docker**, allowing consistent deployment across different environments. **GitHub Actions** is utilized to automate build, testing, and deployment processes through Continuous Integration and Continuous Deployment (CI/CD) pipelines. Finally, the system is deployed on **Microsoft Azure**, providing cloud-based hosting, scalability, reliability, and high availability for enterprise-level operations.

CHAPTER 8: DATABASE DESIGN

Tables:

Employees

Column	Type
EmployeeId	int
Name	varchar
Email	varchar
Department	varchar
Position	varchar

LeaveRequests

Column	Type
LeaveId	int
EmployeeId	int
StartDate	datetime
EndDate	datetime
Status	varchar

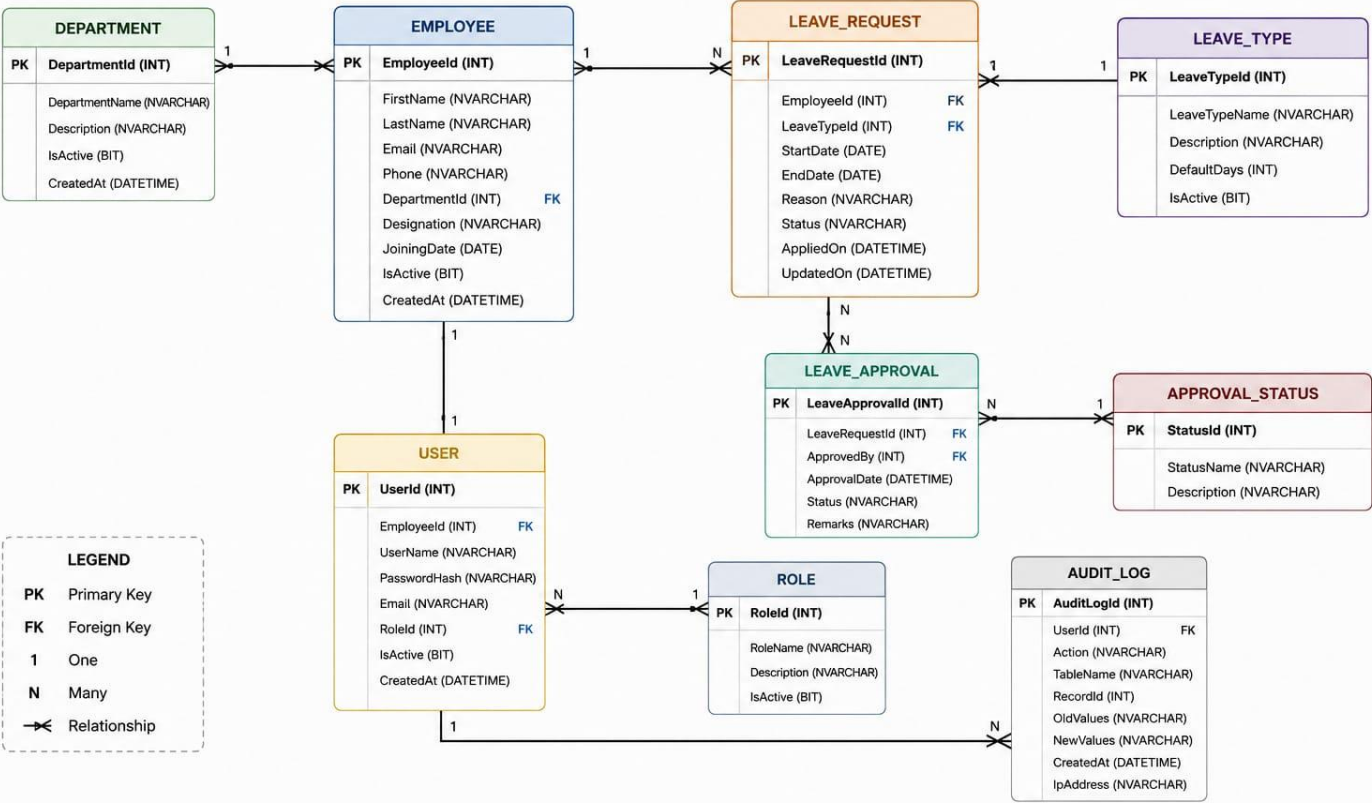
Roles

Column	Type
RoleId	int
RoleName	varchar

ER Diagram:

ER DIAGRAM

HR MANAGEMENT & WORKFORCE ANALYTICS



CHAPTER 9: EMPLOYEE MODULE

- Features:
- Add Employee
 - Edit Employee

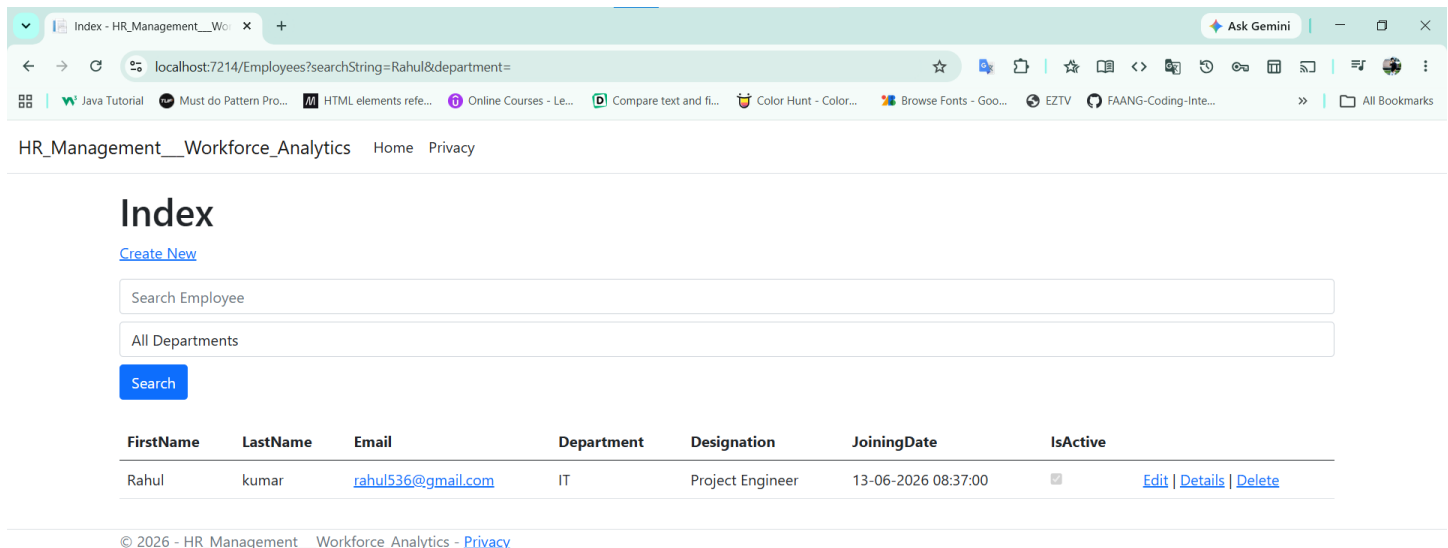
- Delete Employee
- Search Employee
- Department Allocation

Workflow:

HR → Create Employee → Save Database → Display Dashboard

Code Example:

```
public async Task CreateEmployee(Employee employee)
{
    _context.Employees.Add(employee);
    await _context.SaveChangesAsync();
}
```



HR_Management__Workforce_Analytics Home Privacy

Index

[Create New](#)

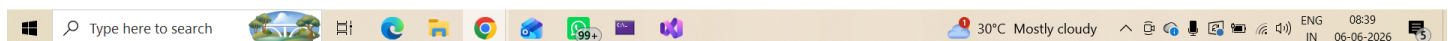
Search Employee

All Departments

[Search](#)

FirstName	LastName	Email	Department	Designation	JoiningDate	IsActive
Rahul	kumar	rahul536@gmail.com	IT	Project Engineer	13-06-2026 08:37:00	☐ Edit Details Delete

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Windows taskbar showing various application icons, system clock, and weather information.

CHAPTER 10: LEAVE MANAGEMENT MODULE

Features:

- Apply Leave
- Approve Leave
- Reject Leave

• Leave History

Workflow:
Employee → Submit Leave → Manager Approval → Database Update
Status Flow:
Pending → Approved
Pending → Rejected

Index - HR_Management__Wo... x +

Ask Gemini

localhost:7214/LeaveRequests?status=Rejected

Java Tutorial Must do Pattern Pro... HTML elements refe... Online Courses - Le... Compare text and fi... Color Hunt - Color... Browse Fonts - Goo... EZTV FAANG-Coding-Inte... All Bookmarks

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Index

Create New

All Status

Filter

EmployeeId	Employee	LeaveType	StartDate	EndDate	Reason	Status
6		casual	19-06-2026 08:49:00	26-06-2026 08:49:00	Going Somewhere	Rejected Edit Details Delete Approve Reject
7		personal	11-06-2026 08:50:00	13-06-2026 08:50:00	unable to talk	Rejected Edit Details Delete Approve Reject

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Type here to search

Very humid 08:53 06-06-2026

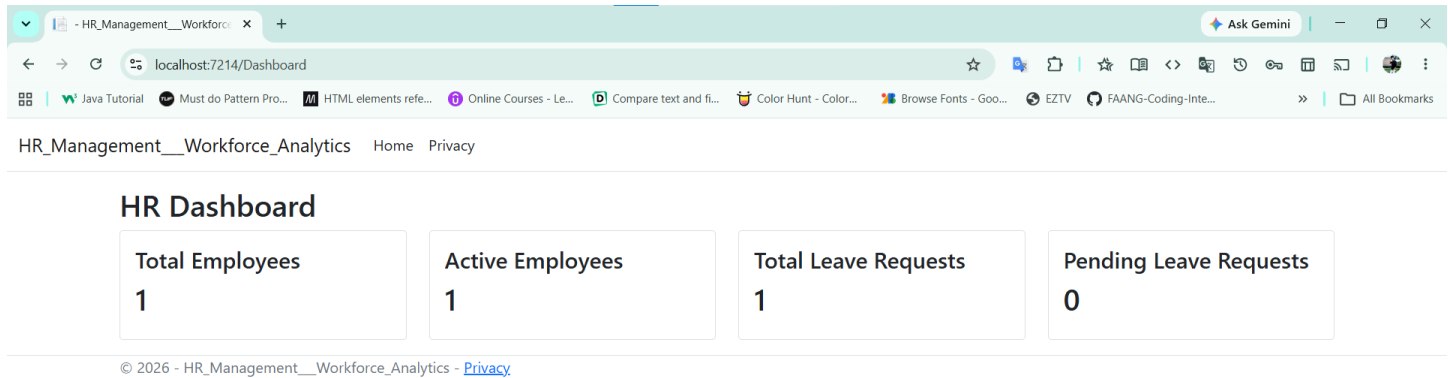
CHAPTER 11: WORKFORCE ANALYTICS DASHBOARD

Dashboard Metrics:

- Total Employees
- Active Employees
- Department Distribution
- Leave Statistics

Charts:

- Pie Chart
- Bar Chart
- Trend Analysis



Benefits:

- Decision Support
- Workforce Monitoring
- Productivity Insights

CHAPTER 12: JWT AUTHENTICATION

Introduction

Security is a critical requirement for any enterprise application, especially when handling sensitive employee information. The HR Management & Workforce Analytics System uses **JSON Web Token (JWT) Authentication** to provide secure access to APIs and application resources. JWT is a compact and self-contained token that securely transmits user identity and authorization information between the client and server.

Purpose

The primary purpose of JWT Authentication is to ensure that only authorized users can access protected resources within the application. It provides secure API access, user authentication, and role-based authorization while maintaining a stateless architecture.

Authentication Process

The JWT authentication workflow follows these steps:

1. **User Login:** The user enters valid login credentials.
2. **Credential Validation:** The server verifies the username and password against the database.
3. **JWT Token Generation:** Upon successful authentication, a JWT token containing user claims and roles is generated.
4. **Token Transmission:** The token is sent to the client application.
5. **Authorization:** The client includes the token in subsequent API requests, and the server validates the token before granting access.

Sample Configuration

In ASP.NET Core 8 MVC, JWT authentication is configured using authentication middleware and security settings. The application registers JWT services during startup using methods such as:

```
builder.Services.AddAuthentication();
```

Additional configurations include defining the secret key, issuer, audience, and token validation parameters.

Advantages of JWT Authentication

Stateless Authentication

JWT does not require session data to be stored on the server, reducing server overhead and improving performance.

Enhanced Security

Tokens are digitally signed, ensuring data integrity and preventing unauthorized modifications.

Scalability

Since authentication information is stored within the token itself, the system can easily scale across multiple servers and cloud environments.

Role-Based Access Control

JWT supports user roles and claims, allowing different access permissions for administrators, HR managers, and employees.

Improved Performance

Token validation is faster than maintaining server-side sessions, resulting in efficient request processing.

Conclusion

JWT Authentication provides a secure, scalable, and efficient mechanism for protecting application resources. By implementing JWT in the HR Management & Workforce Analytics System, user identity is verified, unauthorized access is prevented, and secure communication between clients and APIs is ensured. This enhances overall system security while supporting modern cloud-based and distributed application architectures.

CHAPTER 13: ROLE-BASED AUTHORIZATION

Introduction

Role-Based Authorization is a security mechanism that restricts access to system resources based on the role assigned to a user. In the HR Management & Workforce Analytics System, role-based authorization ensures that users can only access the features and data relevant to their responsibilities. This improves security, protects sensitive information, and prevents unauthorized operations within the application.

User Roles

The system defines three primary roles:

1. Administrator (Admin)

The Administrator has complete control over the system. This role is responsible for managing users, assigning roles, configuring system settings, monitoring activities, and maintaining overall system security.

2. HR Manager

The HR Manager is responsible for managing employee records, approving leave requests, viewing reports, and monitoring workforce-related activities. HR Managers have access to HR functionalities but cannot modify critical system configurations.

3. Employee

Employees have limited access and can perform self-service operations such as viewing personal information, applying for leave, checking leave status, and updating basic profile details.

Permission Matrix

Role	Access Level
Admin	Full Access to All Modules
HR Manager	Manage Employees, Leave Requests, and Reports
Employee	Self-Service Access Only

Implementation

Role-based authorization is implemented using ASP.NET Core's authorization framework. Access to controllers and actions is restricted through role attributes. For example:

```
[Authorize(Roles = "Admin")]
```

This ensures that only users assigned the **Admin** role can access specific resources or functionalities.

Benefits of Role-Based Authorization

- Enhances application security.
- Prevents unauthorized access to sensitive data.
- Simplifies user management.
- Enforces organizational access policies.
- Provides better control over system operations.
- Supports secure and scalable enterprise applications.

Conclusion

Role-Based Authorization plays a crucial role in securing the HR Management & Workforce Analytics System. By assigning permissions based on user roles, the system ensures that each user can access only the features required for their job responsibilities, thereby improving security, accountability, and operational efficiency.

CHAPTER 14: REST APIs & SWAGGER

API Endpoints:

GET /api/employees
POST /api/employees
PUT /api/employees/{id}
DELETE /api/employees/{id}

Swagger Benefits:

- Documentation

-
- The screenshot displays the Swagger UI for a REST API. The top navigation bar shows the URL `localhost:7214/Swagger/index.html`. The main area is titled `GET /api/EmployeeApi`. Below this, there is a **Parameters** section with a `No parameters` message and a `Cancel` button. An `Execute` button is present to test the endpoint. The **Responses** section shows a `200` status code and a `Response body` containing a JSON object:
- ```
{
 "employeeId": 2,
 "firstName": "Sagar",
 "lastName": "Kumar",
 "email": "sagar123@gmail.com",
 "department": "CSE",
 "designation": "Project Engineer"
}
```
- The bottom of the image shows a Windows taskbar with various application icons and a system tray displaying the date and time as 01:39 on 06-06-2026.

Swagger UI

localhost:7214/swagger/index.html

name Description

id \* required  
integer(\$int32) 2  
(path)

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'https://localhost:7214/api/EmployeeApi/2' \
-H 'accept: */*' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJodHRwOi8vc2NoZWl1cy54bWVzbnZmLnR5cy8yZy93cy8yMDAxLzA1L2lkZW50aXR5L2NoYW11cy9uYXV1IiwiaXNjaW50eSI6IjE5ODQ0MjE0IiwiaWF0IjoiMTYxMjE0MjE0In0='
```

Request URL

https://localhost:7214/api/EmployeeApi/2

Server response

Code Details

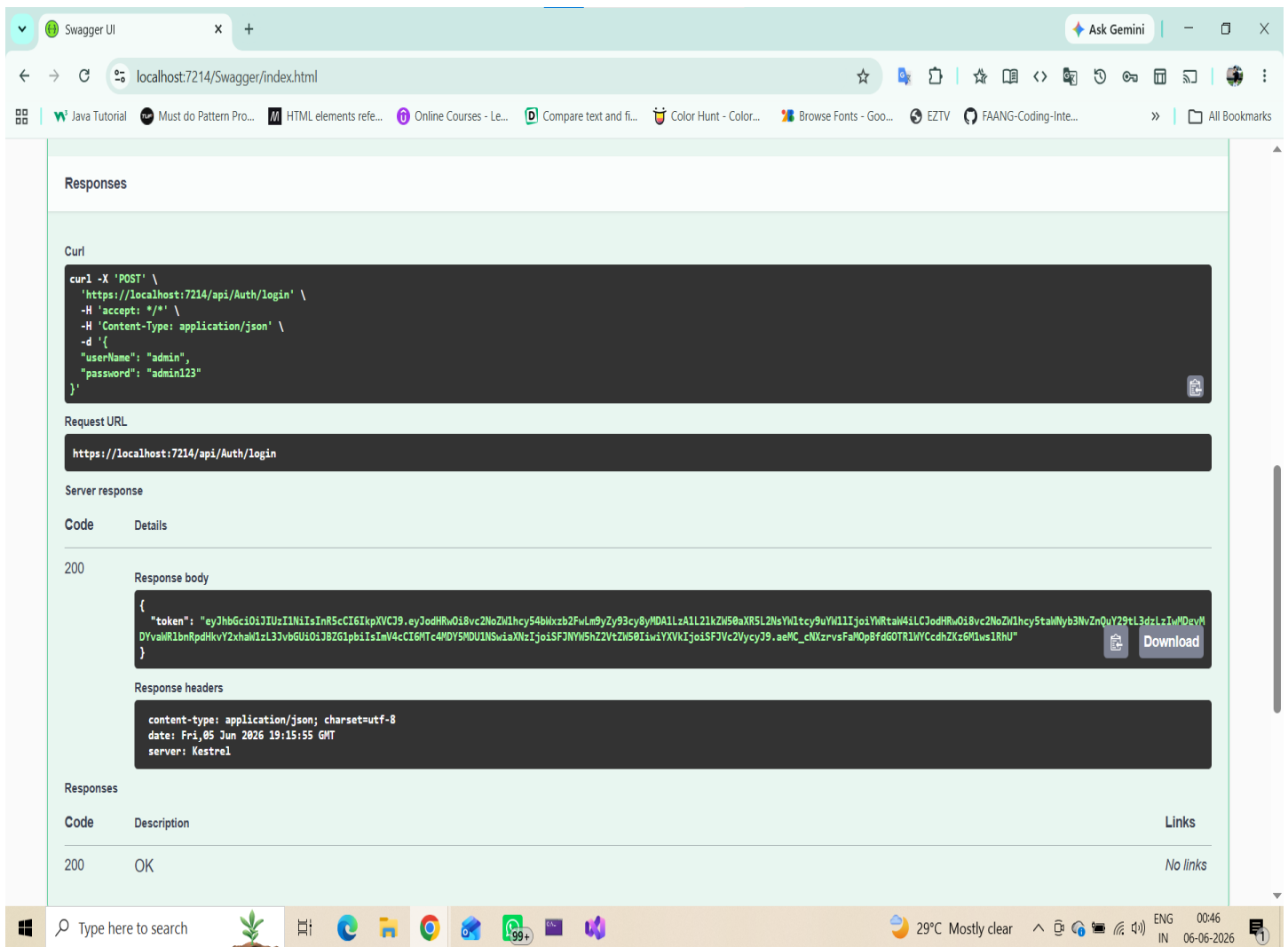
200

Response body

```
{
 "employeeId": 2,
 "firstName": "Sagar",
 "lastName": "Kumar",
 "email": "sagar123@gmail.com",
 "department": "CSE",
 "designation": "Project Engineer",
 "joiningDate": "2026-06-17T02:52:00",
 "isActive": true
}
```

Download





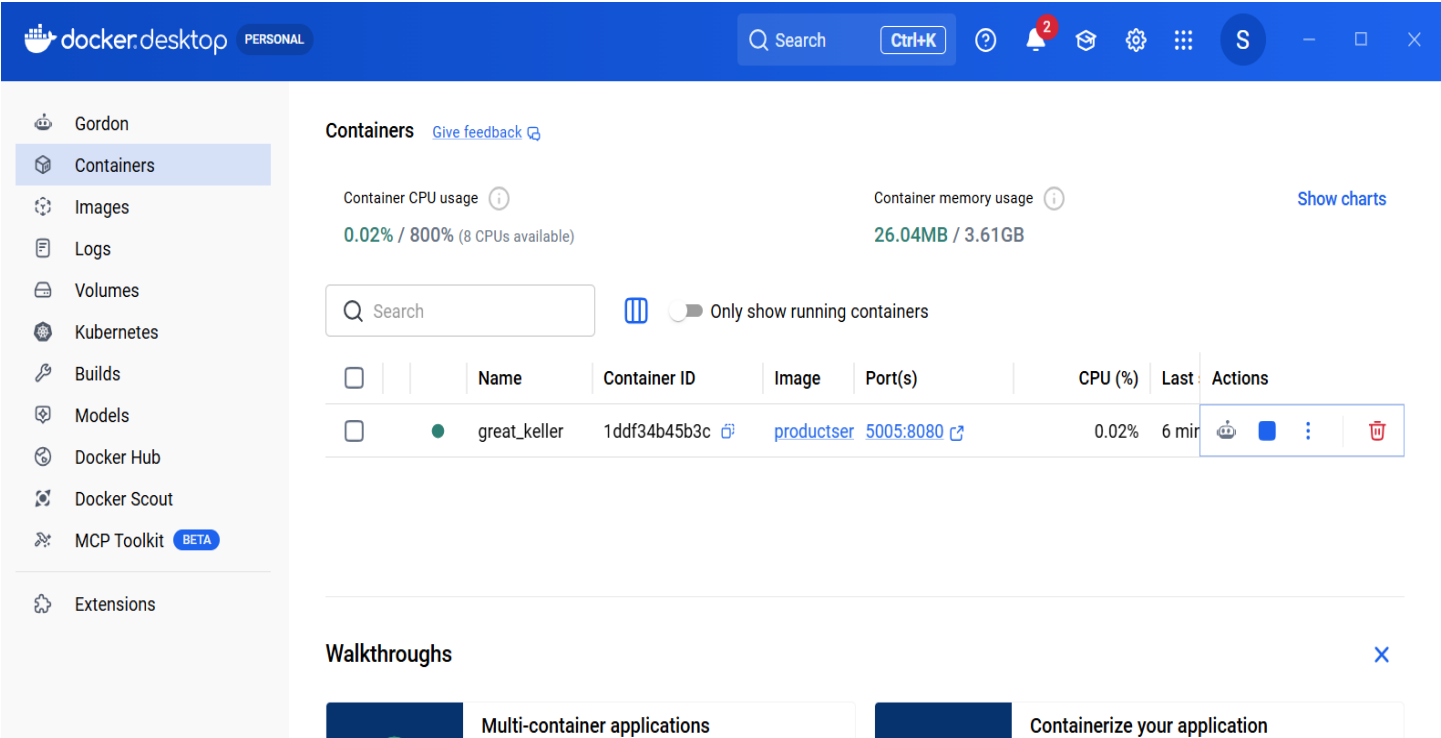
Purpose:  
Containerized Deployment

Dockerfile Components:

- 1. SDK Image
- 2. Build Stage
- 3. Runtime Stage
- 4.

Benefits:

- Portability
- Consistency
- Scalability



CHAPTER 16: GITHUB ACTIONS CI/CD

Pipeline Stages:

Git Push



Build



Test



Docker Build



Azure Deploy

Sample Workflow:

name: CI-CD

on:

push:

branches:

- main

Benefits:

- Automation
- Faster Delivery
- Reduced Errors

The screenshot displays the GitHub Actions interface for a workflow named 'Wipro\_preskilling\_Training'. The workflow is triggered by a push to the 'main' branch. The 'test' job, which runs on a 'ubuntu-latest' runner, has completed successfully in 13 seconds. The workflow file 'ci-cd.yml' is shown, and the annotations section indicates a warning about deprecated actions.

CI-CD Pipeline

✓ Added Github Actions workflow #12

Summary

All jobs

test

Run details

Usage

Workflow file

Triggered via push 4 minutes ago

sagarkumar2301 pushed · cefea43 main

Status: Success

Total duration: 17s

Artifacts: -

ci-cd.yml

on: push

test 13s

Annotations

1 warning

test

Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v4, actions/setup-node@v4. Acti...

## **CHAPTER 17: AZURE DEPLOYMENT**

Azure Services Used:

1. Azure App Service
2. Azure SQL Database
3. Azure Container Registry

Deployment Flow:

GitHub

↓

Azure Pipeline

↓

Azure App Service

Benefits:

- High Availability
- Scalability
- Monitoring



## CHAPTER 18: TESTING

Testing Types:

1. Unit Testing
2. Integration Testing
3. System Testing
4. User Acceptance Testing

Sample Test Cases:

Test Case

Login

Add Employee

Apply Leave

Expected Result

Success

Saved

Created

Command Prompt

```
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Moq' into project 'C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests\HRManagement.Tests.csproj'.
info : GET https://api.nuget.org/v3/registration5-gz-semver2/moq/index.json
info : OK https://api.nuget.org/v3/registration5-gz-semver2/moq/index.json 1155ms
info : Restoring packages for C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests\HRManagement.Tests.csproj...
info : GET https://api.nuget.org/v3-flatcontainer/moq/index.json
info : OK https://api.nuget.org/v3-flatcontainer/moq/index.json 522ms
info : GET https://api.nuget.org/v3-flatcontainer/moq/4.20.72/moq.4.20.72.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/moq/4.20.72/moq.4.20.72.nupkg 191ms
info : GET https://api.nuget.org/v3-flatcontainer/castle.core/index.json
info : OK https://api.nuget.org/v3-flatcontainer/castle.core/index.json 395ms
info : GET https://api.nuget.org/v3-flatcontainer/castle.core/5.1.1/castle.core.5.1.1.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/castle.core/5.1.1/castle.core.5.1.1.nupkg 143ms
info : GET https://api.nuget.org/v3-flatcontainer/system.diagnostics.eventlog/index.json
info : OK https://api.nuget.org/v3-flatcontainer/system.diagnostics.eventlog/index.json 55ms
info : GET https://api.nuget.org/v3-flatcontainer/system.diagnostics.eventlog/6.0.0/system.diagnostics.eventlog.6.0.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/system.diagnostics.eventlog/6.0.0/system.diagnostics.eventlog.6.0.0.nupkg 152ms
info : Installed Moq 4.20.72 from https://api.nuget.org/v3/index.json to C:\Users\ASUS\.nuget\packages\moq\4.20.72 with content hash EA55cJyNn8eTMwgrdZJH5QLFp2L43ox1t1koYUKIE9pRwL7840WitXcCV5ApS+AMYEAlt7Fo03A2XfouVHmQ==.
info : Installed Castle.Core 5.1.1 from https://api.nuget.org/v3/index.json to C:\Users\ASUS\.nuget\packages\castle.core\5.1.1 with content hash rpyTiczKzGpf+EkZgDr9CCITdemhsrwA/W5hMoPjLkRfXzH44zDL0ovXeKtmxblykXX9aJV0DSp1m18CTw2g==.
info : Installed System.Diagnostics.EventLog 6.0.0 from https://api.nuget.org/v3/index.json to C:\Users\ASUS\.nuget\packages\system.diagnostics.eventlog\6.0.0 with content hash lcyUiXTsETK2ALsZrX+nWuHSIQeazhqPhLfaRxzdGaG93+0kELpgEHTwM0lQe7+jSFnKwACAgL4kjeZCQJnw==.
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.06.06.05.55.24/vulnerability.base.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.06.06.05.55.24/2026.06.06.05.55.24/vulnerability.update.json
info : Package 'Moq' is compatible with all the specified frameworks in project 'C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests\HRManagement.Tests.csproj'.
info : PackageReference for package 'Moq' version '4.20.72' added to file 'C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests\HRManagement.Tests.csproj'.
info : Writing assets file to disk. Path: C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests\obj\project.assets.json
log : Restored C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests\HRManagement.Tests.csproj (in 3.35 sec).

C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests>dotnet test
Restore complete (1.4s)
 HRManagement.Tests net10.0 succeeded (16.7s) -> bin\Debug\net10.0\HRManagement.Tests.dll
[xUnit.net 00:00:00.01] xUnit.net VSTest Adapter v3.1.4+50e68bbb8b (64-bit .NET 10.0.8)
[xUnit.net 00:00:01.79] Discovering: HRManagement.Tests
[xUnit.net 00:00:01.97] Discovered: HRManagement.Tests
[xUnit.net 00:00:02.06] Starting: HRManagement.Tests
[xUnit.net 00:00:02.32] Finished: HRManagement.Tests
 HRManagement.Tests test net10.0 succeeded (6.9s)

Test summary: total: 1, failed: 0, succeeded: 1, skipped: 0, duration: 6.8s
Build succeeded in 27.2s

C:\Wipro_preskilling_Training\Final_Capstone_Project_By_Sagar_Kumar\HR Management & Workforce Analytics\HRManagement.Tests>
```

Type here to search



31°C Light rain



ENG 13:58

## CHAPTER 19: RESULTS

### *Introduction*

The **HR Management & Workforce Analytics System** was successfully designed, developed, tested, and deployed using modern web technologies and DevOps practices. The system achieved all the planned objectives and provides a secure, scalable, and user-friendly platform for managing HR operations. Various modules were implemented and integrated to automate employee management, leave processing, authentication, reporting, and deployment activities.

### *Implemented Features*

#### **Employee Management**

The system allows administrators and HR managers to create, update, view, and manage employee records efficiently through a centralized platform.

#### **Leave Management**

Employees can submit leave requests online, while HR managers can review, approve, or reject requests, reducing manual paperwork and processing time.

#### **Analytics Dashboard**

Interactive dashboards provide workforce insights, employee statistics, leave trends, and organizational metrics that support better decision-making.

#### **JWT Security**

JWT-based authentication and authorization ensure secure access to application resources and protect sensitive employee information from unauthorized access.

#### **Swagger APIs**

Swagger documentation enables easy testing, validation, and integration of REST APIs while improving developer productivity.

#### **Docker Implementation**

The application has been containerized using Docker, ensuring consistency across development, testing, and production environments.

#### **GitHub Actions CI/CD**

Automated build and deployment pipelines have been implemented using GitHub Actions, reducing manual deployment effort and improving software delivery efficiency.

#### **Azure Deployment**

The system is deployed on Microsoft Azure, providing high availability, scalability, and reliable cloud-based hosting.

### *Performance Evaluation*

The developed system was tested under different usage scenarios and demonstrated satisfactory performance.

#### **Fast Response Time**

Optimized database queries and efficient backend processing ensure quick response times for user requests and system operations.

## Secure Access

JWT authentication and role-based authorization provide a secure environment for accessing application resources and sensitive data.

## Reliable Deployment

The combination of Docker, GitHub Actions, and Azure ensures stable deployments, minimal downtime, and simplified maintenance.

## Overall Outcome

The project successfully meets its objectives by providing a modern HR management solution that automates administrative tasks, improves data accuracy, enhances security, and delivers valuable workforce analytics. The integration of cloud technologies and DevOps practices further improves system reliability, scalability, and maintainability, making the solution suitable for real-world organizational use.

## CHAPTER 20: FUTURE ENHANCEMENTS

1. Introduction
2. The current version of the **HR Management & Workforce Analytics System** successfully automates employee management, leave processing, authentication, reporting, and deployment. However, there are several opportunities to further enhance the system by introducing advanced HR functionalities, intelligent analytics, and mobile accessibility. These enhancements will improve user experience, increase operational efficiency, and provide deeper business insights.

### *3. Proposed Future Enhancements*

#### **4.1. Attendance Tracking System**

5. A dedicated attendance management module can be integrated to track employee check-in and check-out times, working hours, overtime, and attendance records. This feature will reduce manual attendance tracking and improve workforce monitoring.

## **6.2. Payroll Management Module**

7. The system can be extended to include payroll processing capabilities. This module would automate salary calculations, tax deductions, bonuses, allowances, and payslip generation, reducing administrative effort and improving accuracy.

## **8.3. Performance Management System**

9. A performance evaluation module can be developed to monitor employee goals, achievements, appraisals, and feedback. This will help organizations measure productivity and support employee growth and development.

## **10. 4. AI-Powered Workforce Analytics**

11. Artificial Intelligence and Machine Learning techniques can be integrated to provide predictive analytics, employee attrition analysis, workforce forecasting, and intelligent decision support. These insights can help management make data-driven strategic decisions.

## **12. 5. Mobile Application Development**

13. A mobile application for Android and iOS platforms can be developed to allow employees and managers to access HR services anytime and anywhere. Features such as leave applications, notifications, profile management, and dashboard access can be provided through mobile devices.

## **Power BI Integration**

14. Integration with Microsoft Power BI can provide advanced reporting and data visualization capabilities. Interactive dashboards and business intelligence reports will help management analyze workforce trends, employee performance, and organizational metrics more effectively.

## *Conclusion*

The proposed future enhancements will transform the HR Management & Workforce Analytics System into a comprehensive enterprise HR solution. By incorporating attendance management, payroll processing, performance tracking, AI-driven analytics, mobile accessibility, and Power BI integration, the system can provide greater automation, improved decision-making, and enhanced user experience while meeting the evolving needs of modern organizations.

## CHAPTER 21: CONCLUSION

The **HR Management & Workforce Analytics System** has been successfully designed and implemented to address the challenges associated with traditional HR management processes. The system automates key HR functions such as employee management, leave management, authentication, and workforce reporting, thereby reducing manual effort, improving operational efficiency, and enhancing data accuracy. Through the implementation of role-based authorization and JWT authentication, the application ensures secure access to sensitive organizational information.

This project demonstrates the practical application of modern software development technologies, including **ASP.NET Core 8 MVC**, **C#**, **SQL Server**, **Entity Framework Core**, **Docker**, **GitHub Actions**, and **Microsoft Azure**. The integration of DevOps practices and cloud deployment has improved the scalability, reliability, and maintainability of the application while enabling efficient deployment and management.

Furthermore, the analytics dashboard provides valuable workforce insights that support informed decision-making and strategic planning. The system is designed with a modular architecture, making it flexible for future enhancements such as payroll processing, attendance tracking, AI-driven analytics, and mobile application support.

Overall, the HR Management & Workforce Analytics System successfully achieves its objectives by delivering a secure, scalable, user-friendly, and enterprise-ready solution that meets the evolving needs of modern organizations and contributes to digital transformation in human resource management.

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**"Success is the sum of small efforts, repeated day in and day out."**

— Robert Collier